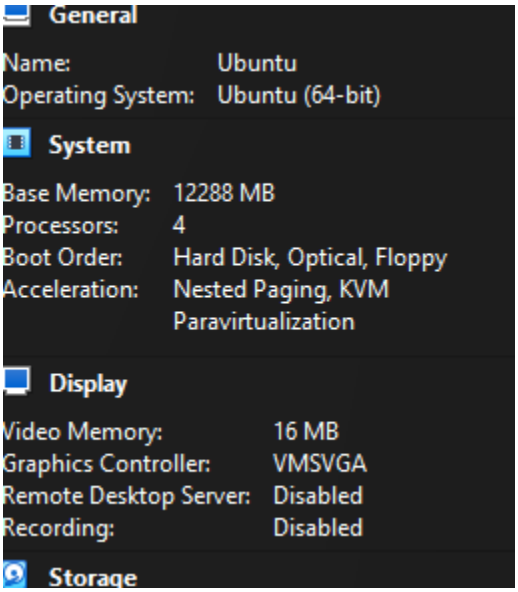
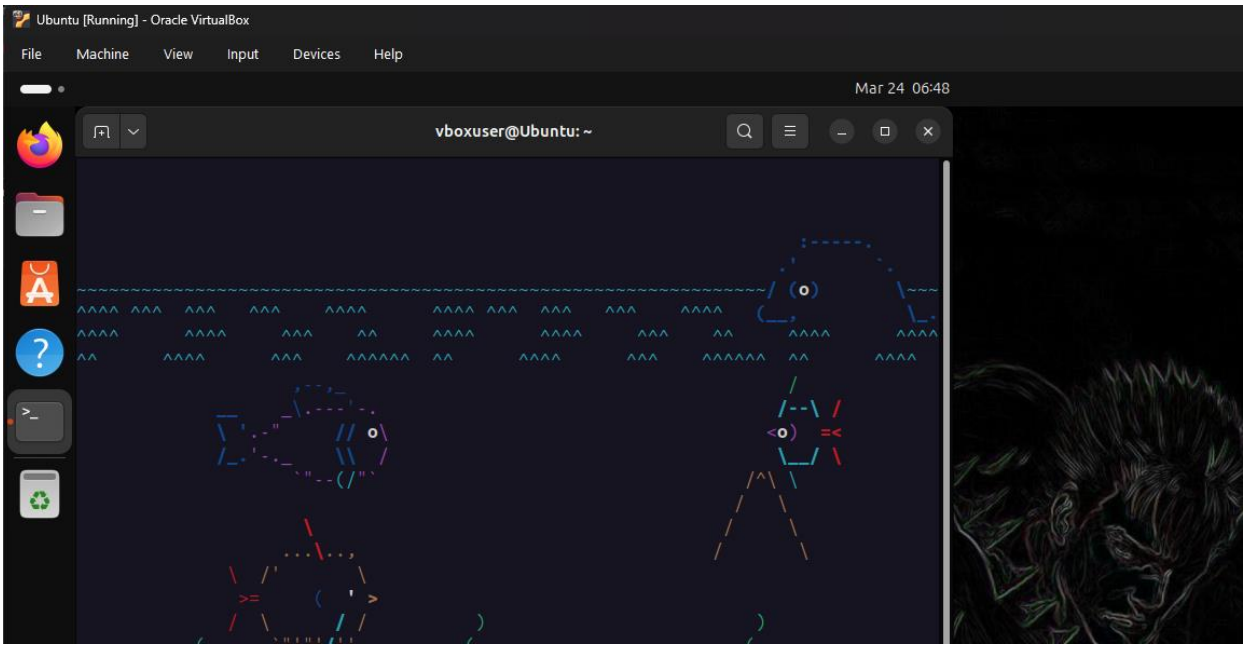
Docker Tasks - 01

- Setup a virtual machine using Ubuntu image.





- Install docker on ec2.

```
[ec2-user@ip-172-31-24-34 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-24-34 ~]$
```

- Pull 5 docker images. (nginx, Apache tomcat, ubuntu, Jenkins, SonarQube)

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```
[ec2-user@ip-172-31-24-34 ~]$ docker run -d -p 8081:80 --name nginx nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
ec781dee3f47: Pull complete
980067d12da2: Pull complete
4174e33a2c9e: Pull complete
6b40784e4837: Pull complete
f0b77348d9b0: Pull complete
0289d65812c3: Pull complete
9baba07a35b6: Pull complete
Digest: sha256:dec7a90bd0973b076832dc56933fe876bc014929e14b4ec49923951405370112
Status: Downloaded newer image for nginx:latest
c9449ae5f2bcac907bd8952db851ef623ff71c7f6c0db218e57391892150b245
[ec2-user@ip-172-31-24-34 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                   NAMES
c9449ae5f2bc   nginx     "/docker-entrypoint.…"   8 seconds ago   Up 6 seconds   0.0.0.0:8081→80/tcp, :::8081→80/tcp     nginx
```

```
[ec2-user@ip-172-31-24-34 ~]$ docker run -d -p 8082:8080 --name tomcat tomcat
Unable to find image 'tomcat:latest' locally
latest: Pulling from library/tomcat
817807f3c64e: Pull complete
de9be28b9519: Pull complete
c318c44e952a: Pull complete
4f4fb700ef54: Pull complete
cc1e0a391268: Pull complete
3adf9b5baee6: Pull complete
5c5afa59de0e: Pull complete
Digest: sha256:6f7e8211ca2c81352a5646ab3a369d16b22e9580fd388c946cc1151d3686fab3
Status: Downloaded newer image for tomcat:latest
dc48353e64a8b405914b4b2b98305859645c1ccbba35f56bf932e389f7a8fcb6
[ec2-user@ip-172-31-24-34 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED              STATUS              PORTS                                   NAMES
dc48353e64a8   tomcat    "catalina.sh run"        5 seconds ago        Up 1 second         0.0.0.0:8082→8080/tcp, :::8082→8080/tcp   tomcat
c9449ae5f2bc   nginx     "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:8081→80/tcp, :::8081→80/tcp       nginx
```

```
[ec2-user@ip-172-31-24-34 ~]$ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
817807f3c64e: Already exists
Digest: sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
[ec2-user@ip-172-31-24-34 ~]$
```

```
[ec2-user@ip-172-31-24-34 ~]$ docker run -d -p 8083:8080 --name jenkins jenkins/jenkins
Unable to find image 'jenkins/jenkins:latest' locally
latest: Pulling from jenkins/jenkins
53c88f1dfeb7: Pull complete
60c0fd0c43b2: Pull complete
e7bfa593efe5: Pull complete
8f0f48748452: Pull complete
9db1c2c6156c: Pull complete
0e3619e74fcd: Pull complete
ea260a68de0f: Pull complete
93325a7fba3a: Pull complete
2327fa553697: Pull complete
01d644112a1c: Pull complete
272c4013e30c: Pull complete
36911cede035: Pull complete
Digest: sha256:5036cab1350e6f10a6fa7865d71369db5a505fcd0507b5b97fde8f97cb1502d9
Status: Downloaded newer image for jenkins/jenkins:latest
dc5689b0940e65af967dfa56dfe1d4fd0ebcf707844c18935534f016b35666fe
[ec2-user@ip-172-31-24-34 ~]$ docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED              STATUS              PORTS                                                   NAMES
dc5689b0940e   jenkins/jenkins   "/usr/bin/tini -- /u…"   6 seconds ago        Up 2 seconds        50000/tcp, 0.0.0.0:8083→8080/tcp, :::8083→8080/tcp   jenkins
dc48353e64a8   tomcat            "catalina.sh run"        About a minute ago   Up About a minute   0.0.0.0:8082→8080/tcp, :::8082→8080/tcp              tomcat
c9449ae5f2bc   nginx             "/docker-entrypoint.…"   3 minutes ago        Up 3 minutes        0.0.0.0:8081→80/tcp, :::8081→80/tcp                  nginx
```

```
[ec2-user@ip-172-31-24-34 ~]$ docker run -d -p 8084:9000 --name sonar sonarqube
Unable to find image 'sonarqube:latest' locally
latest: Pulling from library/sonarqube
817807f3c64e: Already exists
7e9e66015a41: Pull complete
c25811b01351: Pull complete
9b1a2f3bb5f3: Pull complete
91e1054453cc: Pull complete
766c6604a977: Pull complete
8f74db2f15c9: Pull complete
4f4fb700ef54: Pull complete
Digest: sha256:17873f679ac62577ac33deed7b6bd4b08b4154d3711cf55485044d22533e9443
Status: Downloaded newer image for sonarqube:latest
f98710bf807ef855bf22ff016d9968ce3d8bf3f860d4437ec3bc240d291421cd
[ec2-user@ip-172-31-24-34 ~]$ docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED              STATUS              PORTS                                                   NAMES
f98710bf807e   sonarqube         "/opt/sonarqube/dock…"   18 seconds ago       Up 14 seconds       0.0.0.0:8084→9000/tcp, :::8084→9000/tcp                 sonar
dc5689b0940e   jenkins/jenkins   "/usr/bin/tini -- /u…"   About a minute ago   Up About a minute   50000/tcp, 0.0.0.0:8083→8080/tcp, :::8083→8080/tcp   jenkins
dc48353e64a8   tomcat            "catalina.sh run"        3 minutes ago        Up 3 minutes        0.0.0.0:8082→8080/tcp, :::8082→8080/tcp              tomcat
c9449ae5f2bc   nginx             "/docker-entrypoint.…"   4 minutes ago        Up 4 minutes        0.0.0.0:8081→80/tcp, :::8081→80/tcp                  nginx
```

- Run nginx container and expose on port 81.

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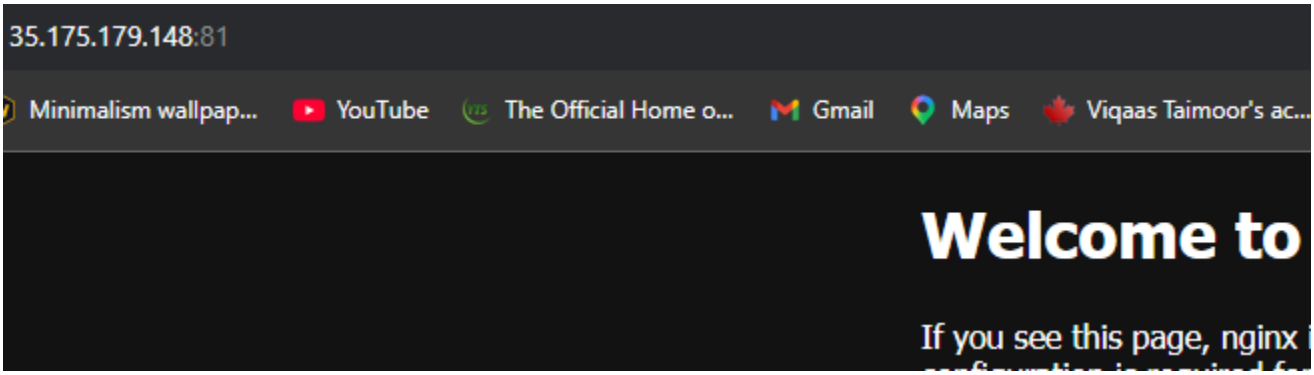
```
[ec2-user@ip-172-31-24-34 ~]$ docker run -d -p 81:80 --name nginx nginx
a345cf68c59d4819ee4ebf1ff3818fab1656554277c0de163b8f70a8b529619a
[ec2-user@ip-172-31-24-34 ~]$ docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED         STATUS         PORTS                                                   NAMES
a345cf68c59d   nginx             "/docker-entrypoint.…"   4 seconds ago   Up 2 seconds   0.0.0.0:81→80/tcp, :::81→80/tcp                         nginx
f98710bf807e   sonarqube         "/opt/sonarqube/dock…"   2 minutes ago   Up 2 minutes   0.0.0.0:8084→9000/tcp, :::8084→9000/tcp                 sonar
dc5689b0940e   jenkins/jenkins   "/usr/bin/tini -- /u…"   3 minutes ago   Up 3 minutes   50000/tcp, 0.0.0.0:8083→8080/tcp, :::8083→8080/tcp      jenkins
dc48353e64a8   tomcat            "catalina.sh run"        5 minutes ago   Up 4 minutes   0.0.0.0:8082→8080/tcp, :::8082→8080/tcp                 tomcat
```



- Delete the Apache Tomcat image from local.

```
[ec2-user@ip-172-31-24-34 ~]$ docker images
REPOSITORY        TAG       IMAGE ID       CREATED       SIZE
tomcat            latest    d10cfd9141f2   3 days ago    417MB
jenkins/jenkins   latest    0a60439ca04e   5 days ago    491MB
sonarqube         latest    92e8e49c2237   7 days ago    1.45GB
nginx             latest    1a1e63136420   7 days ago    161MB
ubuntu            latest    f794f40ddfff   4 weeks ago   78.1MB
[ec2-user@ip-172-31-24-34 ~]$ docker rmi d10cfd9141f2
Untagged: tomcat:latest
Untagged: tomcat@sha256:6f7e8211ca2c81352a5646ab3a369d16b22e9580fd388c946cc1151d3686fab3
Deleted: sha256:d10cfd9141f2060b46e81061e43958ad964c61f345634be6c4143dfb114d01f1
Deleted: sha256:56842d4f66049c9482f96597539ad012a08367ba1221df41f791648b130da370
Deleted: sha256:ca76377844d00f654113e37bb2ad9fac0751576eefa3cc79937fd1738da218c3
Deleted: sha256:1b522345a8767590e493e43da41f6f9fdb7abeb7b056d67a393a368047943ffd
Deleted: sha256:d4545b7d307e71800ca398a54f56034a40a86f5f8409947732755304d602b59b
Deleted: sha256:e4bc3dab8da9b2150ecf8da06896eb4a3a42e21842089fcac00c1f90a527b131
Deleted: sha256:2ce25c23d0a3917082403730af4992832360ec3747048dc9f0c222610d03662d
Deleted: sha256:f14c51c03bbcba4cdc2830f53ee682fcb23db4e605c97601d889fa3975ab00ce
Deleted: sha256:f18dc84dba37c78c81e08d2cb20c6b440bd48199c15c120d26416da5600c9d4e
```

- Inspect the Jenkins image, SonarQube image.

```
[ec2-user@ip-172-31-24-34 ~]$ docker inspect 0a60439ca04e
[
    {
        "Id": "sha256:0a60439ca04e8fde206bcd16711171431fef82be9cfe661b24880ebb30
        "RepoTags": [
            "jenkins/jenkins:latest"
        ],
        "RepoDigests": [
            "jenkins/jenkins@sha256:5036cab1350e6f10a6fa7865d71369db5a505fcd0507
        ],
        "Parent": "",
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-03-18T13:07:45.858264507Z",
        "Container": "",
        "ContainerConfig": {
            "Hostname": "",
            "Domainname": "",
            "User": "",
            "AttachStdin": false,
            "AttachStdout": false,
```

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```
[ec2-user@ip-172-31-24-34 ~]$ docker inspect 92e
[
    {
        "Id": "sha256:92e8e49c2237cc0dc6c980222212affa648a958d0584f91741
        "RepoTags": [
            "sonarqube:latest"
        ],
        "RepoDigests": [
            "sonarqube@sha256:17873f679ac62577ac33deed7b6bd4b08b4154d371
        ],
        "Parent": "",
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-03-17T02:48:55.794786282Z",
        "Container": "",
        "ContainerConfig": {
            "Hostname": "",
            "Domainname": "",
            "User": "",
            "AttachStdin": false,
            "AttachStdout": false,
            "AttachStderr": false,
            "Tty": false,
```

- Run Jenkins container and run one sample job.

```
[ec2-user@ip-172-31-24-34 ~]$ docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED          STATUS          PORTS                                                                    NAMES
a345cf68c59d   nginx             "/docker-entrypoint.…"   15 minutes ago   Up 15 minutes   0.0.0.0:81→80/tcp, :::81→80/tcp                                          nginx
f98710bf807e   sonarqube         "/opt/sonarqube/dock…"   18 minutes ago   Up 17 minutes   0.0.0.0:8084→9000/tcp, :::8084→9000/tcp                                  sonar
dc5689b0940e   jenkins/jenkins   "/usr/bin/tini -- /u…"   19 minutes ago   Up 19 minutes   50000/tcp, 0.0.0.0:8083→8080/tcp, :::8083→8080/tcp                       jenkins
[ec2-user@ip-172-31-24-34 ~]$
```

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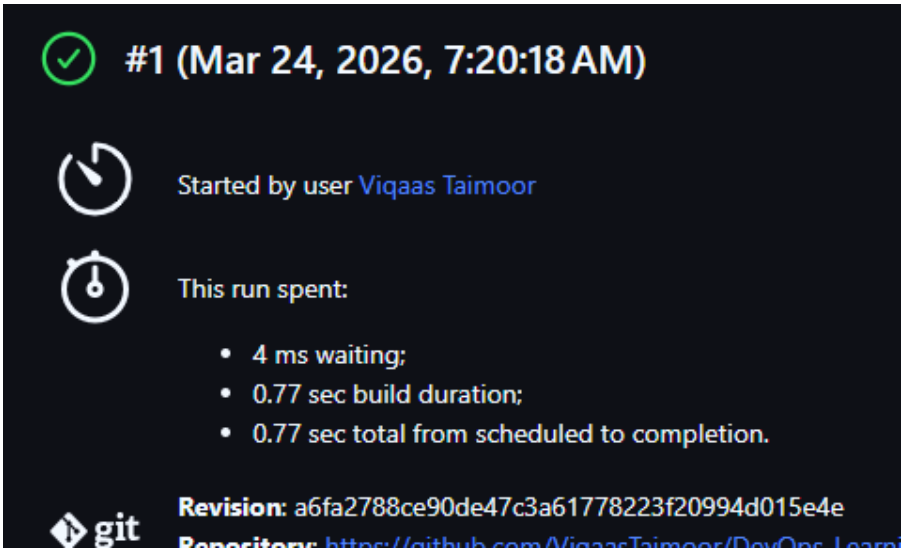

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